

BSc (Hons) Computer Science
BSc (Hons) Cyber Security
Module Code: QH0305 Module Title: Problem Solving
Assessment Sheet 5

Instructions:

This is one of the eight assessment tasks which will contribute to the overall mark. You will need to complete the tasks as outlined below and then document them in a Word document file. As a minimum, you should provide screenshots of the following:

- Your code
- The output that your code generates

In instances where your code could generate different outputs depending on what values are given, you should provide multiple screenshots of the console screen, showing the different outputs in order to demonstrate that the code works correctly.

Your Word document should have appropriate headings to ensure that each task can easily be identified alongside the rest of your work.

This assessment will focus on loops, conditional logic, and calendar formatting.

You must attempt all tasks on this sheet to achieve a higher grade. For example, if you want to gain marks between 70 - 100, you must complete all other grades first and add them to your portfolio with screenshots.

A zip folder with all Grade codes (in .c or .txt format) must be attached inside of the portfolio (MS Word file). This is in addition to the code presented as screenshots.

Assessment Task 5: Temperature Logger

In this task, you will create a program that logs daily temperatures for a given month (up to 30 days). The program will use **arrays** to store temperature data and provide **various menu options** for analysis.

Task

Requirements:

- Prompt the user to enter the **number of days**
- Based on that number, create an **array** to store temperature readings for each day
- Allow the user to input **temperatures** one by one
- Implement a **menu** with the following options:

1. *Display all temperature readings.*
2. *Calculate and display the average temperature overall.*
3. *Find and display the highest and lowest temperature.*
4. *Count the number of days above and below a user-defined threshold.*
5. *Calculate and display the weekly average temperature.*
6. *Reverse the order of temperature readings and display them.*
7. *Exit the program.*

NOTE: Only the final version of the program (marks 70-100) will have all the above menu options implemented. For lower marks, only some options need to work as outlined below.

To achieve Marks (Between 40-49):

- Ask the user for how many days they wish to record temperature readings for.
- Allow user input to store all temperature readings in an array of appropriate size.
- Implement menu options 1 and 2 only.
- Demonstrate user using both options to display all entered temperatures and to calculate the average.
- Submit screenshots of code and outputs.

To achieve Marks (Between 50-59):

- Complete all the previous steps.
- Implement option 3 (calculating the highest and the lowest temperature).
- Implement option 7 (exit).
- Ensure the program runs in a loop until the user exits.
- Handle invalid menu choices gracefully.

To achieve Marks (Between 60-69):

- Complete all the previous steps.
- Refactor your code so that your array is of fixed size (e.g. 50) and your code makes use of a terminator/sentinel value (such as -1) as much as possible.
- Implement option 4 (counting days above/below a threshold). The threshold should be taken from the user, as their input.
- Implement option 5 (calculate the average temperature for each week). Assume that first user input is day 1 of week 1, and so that every 7 entries count as a week.
- Ensure all menu actions function correctly.
- Explain how the program works either via in-code comments or within your portfolio.

To achieve Marks (Between 70-100):

- Complete all the previous steps.
- Implement option 6 (reverse the order of temperature readings and display them). The output should be formatted in the following way:

Today: 29C
Yesterday: 24C
2 days ago: 26C
3 days ago: 28C
4 days ago: 27C
Etc.

(Assuming at the end of array user entered [..., 27,28,26,24,29])

- Research and explain how **pointers** work, and why they are useful in this context.
- Rewrite part of the program using **pointers** instead of array indexing to:
 - Traverse the temperature array
 - Reverse the array
 - Find the highest/lowest temperature

Assignment Preparation Guidelines

- All components of the assignment report must be Word-processed (**handwritten text or hand drawn diagrams are not acceptable**), font size must be within the range of 11 point to 14 point including the headings, body text and any texts within diagrams.
- Standard and commonly used fonts such as Times New Roman, Arial or Calibri should be used.
- All figures, graphs and tables must be numbered and labelled with short explanations.
- Material from external sources must be properly acknowledged and cited within the text using the Harvard referencing system.
- All components of the assignment (text, diagrams, code etc.) and all assessment sheets must be submitted in one Word file.
- The report should be logically structured, the core of the report may start by defining the problem / requirements, followed by the proposed solution including a detailed discussion, analysis and evaluation, leading to the implementation and testing stage, finally a conclusion and personal reflection on learning.
- Screenshots without description / discussion are not suitable as they do not express your understanding or support your work adequately.

Submission instructions

- This is a portfolio assignment with eight tasks in total. Each task will be completed and saved in the portfolio. Once the portfolio is completed, it should be submitted on Turnitin. The submission link to Turnitin can be found under the “Assessment Tab” in your module section in the SOL VLE.
- Please note file size limitation might apply. Your report must be under 250MB.
- The source code for each task should be **zipped** and **attached** to your Word document report submission in the appendix.
- The Assignment Brief can be found under the “Assessment Tab” in your module section in the SOL VLE.
- **Refer to the Assignment Brief** to find the links to Late Submissions, Extenuating Circumstances, Academic Misconduct, Ethics Policy, Grade marking and Guidance for online submission through Solent Online Learning (SOL).